

PROFESSIONAL SUMMARY

B.Tech Computer Science student building deployed Generative AI applications with the Gemini API alongside hand-coded full-stack systems. Experience with structured JSON-mode outputs, Streamlit deployment, and backend engineering (FastAPI, PostgreSQL, Redis, WebSockets, Docker). Strong Python foundation with Pandas/NumPy and REST APIs. Available for the full 2-month internship duration.

TECHNICAL SKILLS

- Generative AI:** Gemini API, prompt engineering, structured/JSON-mode outputs, AI-assisted development workflows
- Programming Languages:** Python (proficient), JavaScript, HTML/CSS, C++, Java (basic)
- Web & App Frameworks:** Streamlit, FastAPI, REST API design, WebSockets, basic frontend (HTML/CSS/JS, Leaflet.js)
- Data & Backend:** PostgreSQL, SQLAlchemy (ORM), Redis, Pandas, NumPy, Docker & Docker Compose
- ML / Data Science:** TensorFlow (basic), Scikit-learn (basic), Jupyter Notebook, Google Colab
- Tools & Platforms:** Git, GitHub, VS Code, Postman, Linux (basic), Vercel / Streamlit Cloud deployment

PROJECTS

- AI Medical Report Analyzer

Python · Streamlit · Gemini API · JSON Mode

 - Built and deployed a Streamlit web app that uses the Gemini API to parse medical reports into clear, plain-language text.
 - Used structured JSON-mode output from Gemini to reliably extract and organize report content before rendering.
 - Generated a clean, readable PDF version of the analyzed report as the final deliverable for the end user.
 - Directly applies the prompt design, JSON-output, and deployment workflow this internship is built around.
- Delhi Metro Live-Tracking Simulator

Python · FastAPI · PostgreSQL · Redis · WebSockets · Docker

 - Designed and hand-coded a full-stack live train tracking system end-to-end: database schema, backend API, real-time simulation, and frontend map.
 - Modeled the system with SQLAlchemy ORM across 5 related tables (lines, stations, trains, trips, interchanges) on PostgreSQL.
 - Built a separate simulator process that computes train movement on a timed loop and writes live positions to Redis.
 - Implemented a FastAPI backend with REST endpoints and a WebSocket stream pushing live position updates to a Leaflet.js map every 10 seconds.
 - Containerized the full stack (API, PostgreSQL, Redis) with Docker and Docker Compose for reproducible deployment.
- Dhaara — AI Traffic Management System

Python · OpenCV · Computer Vision

 - Built a traffic management prototype using Python and OpenCV to analyze live video feeds and estimate vehicle density.
 - Designed adaptive signal-timing logic that adjusts to real-time traffic conditions to reduce congestion.
 - Used AI-assisted development (Gemini) to accelerate iteration while writing and integrating the OpenCV pipeline.

EDUCATION

Qualification	Institution	Year	Score
B.Tech, Computer Science Engineering	Hindustan College of Science & Technology, Mathura	Present	8.24 CGPA
XII (CBSE)	All Saints School, Agra	2024	74.2%
X (CBSE)	All Saints School, Agra	2022	83.2%

ACHIEVEMENTS

- Won Innovation Day at Hindustan College of Science and Technology for applied AI project work.
- Maintained an 8.24 CGPA through the 3rd semester of B.Tech.
- Self-taught Generative AI integration, backend APIs, and data analysis through independent project development.

LANGUAGES

English | Hindi